



UPT-Flow: Multi-scale transformer-guided normalizing flow for low-light image enhancement

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ABSTRACT

Low-light images often suffer from information loss and RGB value degradation due to extremely low or nonuniform lighting conditions. Many existing methods primarily focus on optimizing the appearance distance between the enhanced image and the normal-light image, while neglecting the explicit modeling of information loss regions or incorrect information points in low-light images. To address this, this paper proposes an Unbalanced Points-guided multi-scale Transformer-based conditional normalizing Flow (UPT-Flow) for low-light image enhancement. We design an unbalanced point map prior based on the differences in the proportion of RGB values for each pixel in the image, which is used to modify traditional self-attention and mitigate the negative effects of areas with information distortion in the attention calculation. The Multi-Scale Transformer (MSFormer) is composed of several global-local transformer blocks, which encode rich global contextual information and local fine-grained details for conditional normalizing flow. In the invertible network of flow, we design cross-coupling conditional affine layers based on channel and spatial attention, enhancing the expressive power of a single flow step. Without bells and whistles, extensive experiments on low-light image enhancement, night traffic monitoring enhancement, low-light object detection, and nighttime image segmentation have demonstrated that our proposed method achieves state-of-the-art performance across a variety of real-world scenes. The code and pre-trained models will be available at <https://github.com/NJUPT-IPR-XuLintao/UPT-Flow>.

1. Introduction

The swift advancement in computer vision technology has contributed to the widespread adoption of digital image processing systems into our daily lives, encompassing monitoring systems, medical image processing, face recognition, satellite remote sensing, and more. However, images captured under uncontrollable low-light conditions, characterized by insufficient or unbalanced lighting, limited exposure time, often exhibit uncertain degradation factors, such as low visibility, color degradation, low contrast, and low signal-to-noise ratio (SNR) [1]. Such low-quality images not only make it difficult for people to obtain information, but also result in inaccurate information expression, posing great challenges for advanced computer vision tasks, such as object detection [2], image segmentation [3], and other scene understanding. Therefore, improving the perceptual quality of low-light images holds great significance.

Recently, the low-light image enhancement (LLIE) task has drawn a lot of interest, with many outstanding works being published [4].

Some supervised learning methods [1,5–10] employ aligned paired information to construct per-pixel constraints between outputs and ground truth. However, models trained with L1 loss may exhibit results blurring [11], as shown in Fig. 2(b). Some methods based on zero-shot [12–14] or unsupervised learning [15,16] overcome the limitation of requiring paired datasets for training. Nevertheless, these approaches often come with problems such as insufficient enhancement, strong noise, and color distortion, as shown in Fig. 2(c). LLFlow [17] is the first to introduce the normalizing flow into the LLIE tasks. LLFlow only uses the maximum likelihood probability estimation method instead of L1 loss for network training, resulting in the enhanced images being more delicate and visually pleasing. However, LLFlow fails to consider the different characteristics of different regions in low-light images, resulting in significant disparities in brightness compared to reference images and a decrease in PSNR, as shown in Fig. 2(d).

In this paper, our key insight is that the degree of information distortion in different regions of low-light images often varies significantly.

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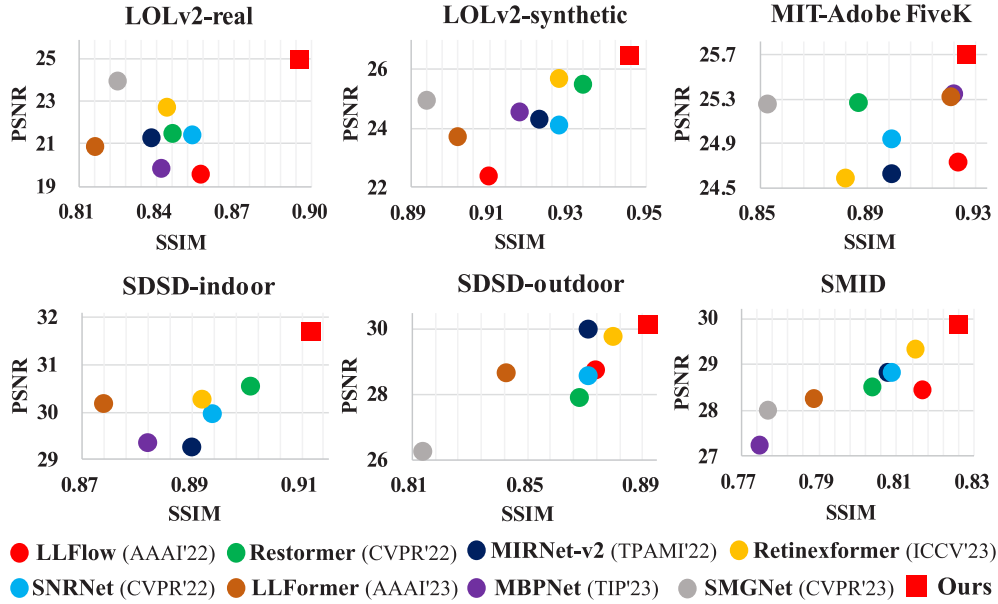


Fig. 1. Our method consistently achieves better performance in terms of PSNR/SSIM over 8 SOTA methods on 6 different benchmarks. Each plot is for comparison on one dataset.

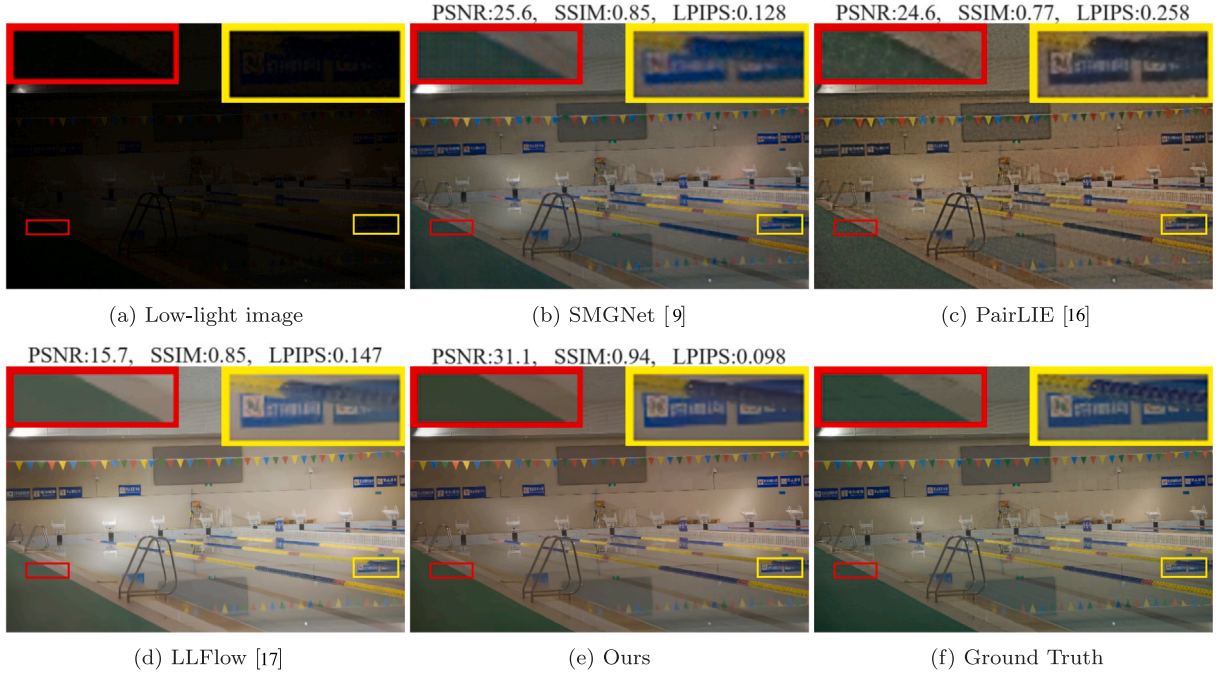


Fig. 2. Examples of the enhanced low-light images and magnified view of the unbalanced point regions.

Some regions with extremely low brightness tend to exhibit severe information loss and color degradation, whereas other areas within the same image may retain acceptable level of visibility. Fig. 3 shows the degradation of RGB components in low-light image and ground truth, although the ratio curve of low-light image shares roughly the same shape as the corresponding ground truth, the difference in some extremely low-light regions is still huge. To enhance the overall image quality effectively, we should adaptively consider different regions in low-light images.

Paper [18] defines some pixels with very low intensity in at least one color (RGB) channel as dark pixels based on statistical data, and thus defines a dark channel prior that can directly provide locations for missing points in images. Paper [19] proposes that low-light images

often have high saturation due to extremely low-light or nonuniform-light, where the RGB three channels contain one or more channels with values of zeros or close to zeros. Inspired by them, we propose an unbalanced point map prior based on the differences in the proportion of RGB components for each pixel in low-light images, as shown in the fourth subplot of Fig. 3. The unbalanced point map includes pixel points where information is severely distorted or lost due to extremely low brightness or uneven lighting. The unbalanced point map helps us construct adaptive attention for different regions in low-light images, achieving the best results in both local details and overall perception, as shown in Fig. 2(e).

The existing conditional normalizing flows [11,17,20,21] suffer from insufficient feature information extraction and limited network

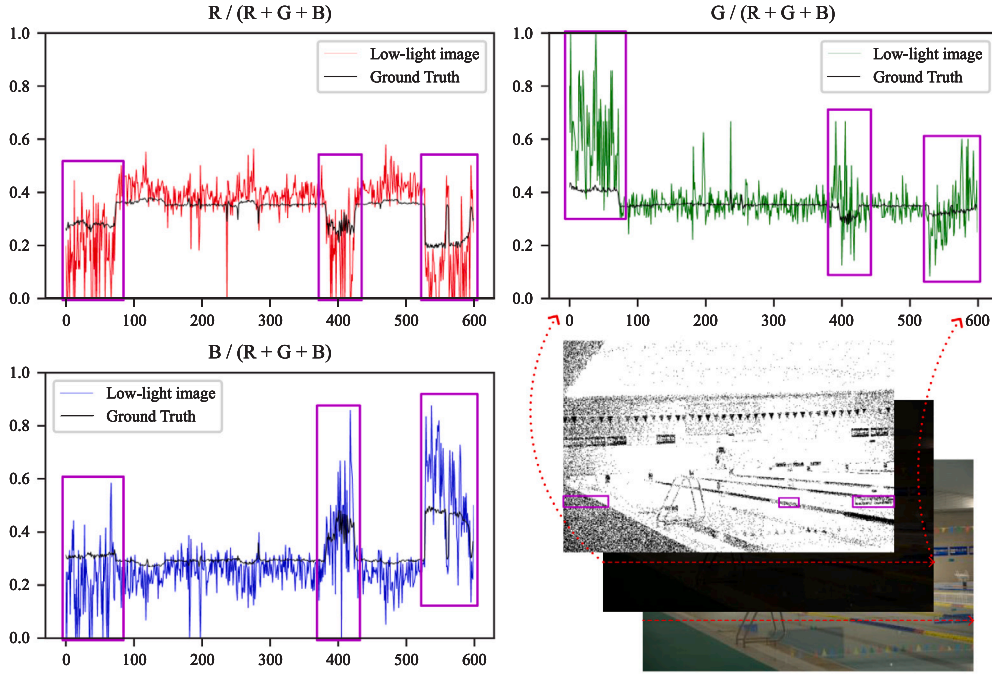


Fig. 3. The RGB proportion of low-light image and ground truth along the red parallel dashed line. The purple rectangle shows the huge gap in RGB composition between low-light image and ground truth in extremely dark region. The binary image in the fourth subplot is our unbalanced point map of low-light image.

expression ability. For this purpose, we propose an Unbalanced Points-guided multi-scale Transformer-based conditional normalizing Flow (UPT-Flow) for low-light image enhancement. The Multi-Scale Transformer (MSFormer) is composed of Window-based RGB value Unbalanced Points-guided Transformer blocks (WUPT) for local distortion information recovery and Global Crossed-channel Feature Transformer blocks (GCFT) for global information compensation. MSFormer utilizes multi-scale features extracted from low-light images to guide conditional normalizing flow for invertible bidirectional mapping learning. Additionally, the conditional affine coupling layer in existing flow models [11,17,20,21] only executes transformation on half of the flow feature, making it difficult to learn complex feature representations. Therefore, we propose alternating cross-coupling between each set of affine layers, enabling all flow features to undergo sufficient feature transformation. Moreover, we have developed a spatial and channel Attention-based Affine Transformation Network (AATN), which helps better fuse the low-light image conditional features extracted by MSFormer with the flow features of normal-light images, enhancing the expressive capability of flow network.

We have conducted extensive experiments on low-light image enhancement, night traffic monitoring image enhancement, low-light object detection, and nighttime semantic segmentation, showing that our proposed method consistently achieved state-of-the-art (SOTA) performance in various scenarios (The quantitative results on low-light image enhancement are previewed in Fig. 1). Moreover, we have conducted a large-scale user study to validate the alignment of our method with human perception. In summary, the contributions of our work are as follows:

- We propose an unbalanced point prior based on the difference in the proportion of RGB three channels and introduce it into the attention calculation of the Transformer.
- We design a U-shaped MSFormer for low-light image encoding for conditional normalizing flow, which fully combines global and local information, providing rich multi-level conditional features for flow.
- In the invertible network of normalizing flow, we develop cross-coupling conditional affine layers based on channel and spatial

attention to efficiently learn flow features, thus enhancing the expressive capability of a single flow step.

- UPT-Flow deeply explores the intrinsic characteristics of low-light images and investigates the unbalanced of RGB pixels in low-light conditions for the first time. It also improves the normalization flow, achieving SOTA performance consistently across multiple benchmarks.

2. Related work

2.1. Low-light image enhancement

2.1.1. Traditional methods

The mathematical model-based approaches were the predominant methods in LLIE tasks before the emergence of data-driven methods, which can be broadly categorized as Histogram Equalization (HE)-based methods and Retinex model-based methods. HE-based methods enhance low-light images by redistributing the distribution of images' pixel value. The basic theory of Retinex theory is to decompose images into two parts: reflectance and illumination, which are individually adjusted to achieve normal exposure images. Guo et al. [22] introduced the illumination map estimation method (LIME), which initializes an illumination map by finding the maximum of three channels of pixels and then imposing a structure prior to it. Hu et al. suggested correcting the color aberration in low-light images by modifying the saturation of HSV-3S [19] or FHSI [23]. Jeon et al. [24] proposed an efficient and fast LLIE method based on an inverted atmospheric scattering model and a pixel-adaptive gamma correction. However, due to the limitations of prior knowledge, the above traditional methods often struggle to adapt to complex scenes, leading to unnecessary distortions.

2.1.2. Deep learning-based methods

The deep learning-based approaches can be primarily categorized into supervised learning, unsupervised learning, and zero-shot learning. With its proposed stacked autoencoder for denoising and augmentation, LLNet [5] is the first study to utilize deep neural networks for LLIE tasks. Wei et al. [25] proposed RetinexNet, which coupled the Retinex

theory with deep networks and denoised the reflectance component using a denoising algorithm [26]. Zhang et al. proposed KinD [27] and KinD++ [28], which were Retinex-based fully-supervised learning models trained under paired images, decomposing the image into its illumination for lighting adjustment and its reflectance for degradation removal. Yang et al. [6] proposed a sparse gradient minimization sub-network for robust LLIE and constructed a large-scale LOf-Light (LOL) paired image dataset, which is the most authoritative database for performance measurement. Zamir et al. proposed MIRNet [8] and MIRNet-v2 [29], which dynamically select the useful set of kernels from multi-scale branches using a self-attention mechanism and preserve the original feature information at each spatial resolution. Xu et al. [1] proposed SNRNet to dynamically enhance the pixels with spatial-varying operations by exploiting long and short-range operations with an SNR prior. Xu et al. [9] proposed SMGNet, which combines a modified generative model for structure modeling and a structure-guided enhancement module. Zhao et al. [30] proposed non-regularization self-supervised Retinex approach simplifies the illumination estimation parameter tuning process by eliminating the need for complex regularization terms. Jiang et al. [31] introduced a Degradation-to-Refinement Generation Network (DRGN), which improves naturalness by decomposing the LLIE into degradation learning and content refinement. Wang et al. [32] decomposed the low-light low-resolution face restoration task into face structural fidelity maintaining and texture consistency learning. Khan et al. [33] introduced the Uneven-to-Enliven network (U2E-Net), which designed to enhance image visibility and color balance under uneven and challenging conditions. In order to perform ultra-high-definition (UHD) LLIE, Wang et al. [10] introduced an axis-based multi-head self-attention and cross-layer attention transformer-based method (LLFormer), which reduces the computational bottleneck for the UHD-LLIE task.

The above-mentioned deep learning-based LLIE methods all require paired normal/low-light images for training. In order to mitigate the dependency of deep models on paired data, in recent years, there has been a rise in the development of unsupervised methods or zero-shot methods. To modify the dynamic range of images, Guo et al. developed ZeroDCE [12] and ZeroDCE++ [13], which estimate a set of curve parameters using deep networks. To concurrently accomplish light estimation and noise removal, Liu et al. [14] developed RUAS, which innovatively enhanced low-light images through a principled framework, integrating the Retinex rule and architecture search for efficient optimization. Long et al. [15] proposed SCI, which created a multi-level lighting learning framework with weight sharing and constructed a self-calibration module within it. Fu et al. [16] developed PairLIE, which uses a solution that combines Retinex and deep networks to learn self-constraints from low-light images, reducing the reliance on manually crafted priors. These methods do not require paired training image samples, greatly improving their universality, but their enhancement results often suffer from detail loss and color degradation due to the lack of reliable guidance.

Despite these advancements, existing methods primarily focus on optimizing the appearance distance between enhanced and normal-light images, neglecting the explicit modeling of information loss regions or incorrect information points, which can lead to detail loss and color degradation. The proposed UPT-Flow designs unbalanced points map for the distortion information in low-light images, explicitly modeling the regions with information distortion, which guides the attention learning of the Transformer. Furthermore, UPT-Flow leverages normalizing flows for exact likelihood computation, effectively capturing fine details in the image data distribution. This precise modeling contrasts with approaches that rely on approximate likelihoods or reconstruction errors, which often lead to blurring and loss of detail. We will compare our method with these excellent methods in subsequent experiments.

2.2. Normalizing flows

Normalizing flows [34–36] are powerful deep generative probabilistic models that excel in efficiently and precisely calculating likelihoods and generating samples, which can bi-directionally transform between complex distributions (e.g., an image) and simple distributions (e.g., Gaussian distribution). Specifically, consider a random variable $\mathbf{Z} \in \mathbb{R}^D$ with a known, manageable probability density function $p_{\mathbf{Z}} : \mathbb{R}^D \rightarrow \mathbb{R}$. Let $\mathbf{g}$ represent an invertible function, and define $\mathbf{Y} = \mathbf{g}(\mathbf{Z})$. Utilizing the change of variables formula, the probability density function $p(\mathbf{y})$ of the random variable $\mathbf{Y}$ can be computed as:

$$\begin{aligned} p(\mathbf{y}) &= p_{\mathbf{Z}}(\mathbf{f}(\mathbf{y}))|\det D\mathbf{f}(\mathbf{y})| \\ &= p_{\mathbf{Z}}(\mathbf{f}(\mathbf{y}))|\det D\mathbf{g}(\mathbf{f}(\mathbf{y}))|^{-1}, \end{aligned} \quad (1)$$

where $\mathbf{f}$ represents the inverse of $\mathbf{g}$ (e.g., $\mathbf{f}^{-1} = \mathbf{g}$), and $D\mathbf{f}(\mathbf{y}) = \frac{\partial \mathbf{f}}{\partial \mathbf{y}}$ denotes the Jacobian of $\mathbf{f}$. The expression (1) enables network training through the minimization of the negative log-likelihood $-\log(p_{\mathbf{Y}}(\mathbf{y}))$.

Recently, normalizing flows have shown its superiority in low-level vision tasks. Winkler et al. [37] proposed conditional normalized flow for super-resolution and vessel segmentation, employing normalization flow to learn the likelihood of conditional distributions. In this setup, supervised prediction tasks can be probabilistically framed. Andreas et al. [11] proposed SRFlow, which tackles the ill-posed nature of the problem by learning the conditional distribution of high-resolution images given low-resolution inputs, achieving diverse and realistic results. HCFlow [20], a unified hierarchical conditional flow framework, combines image super-resolution and rescaling tasks by simultaneously modeling low-resolution images and their high-frequency components, achieving improved visual quality through optimized various losses. Flow-Fill [21] is a two-stage image inpainting framework that employs a conditional normalizing flow model for real-time, diverse results in completing large corrupted areas and enables region-specific semantic transfer tasks.

3. Method

3.1. Overall pipeline

Fig. 4 shows the overview of our framework, primarily comprising a low-light image encoder consisting of a U-shaped Multi-Scale Transformer (MSFormer) and an invertible network of conditional normalizing flow. Given a low-light image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$, we first compute an unbalanced point map based on the characteristics of the pixel value distribution. Subsequently, the feature map of the low-light image and the unbalanced point map pass through a 4-level symmetric MSFormer for deep feature extraction, and multi-scale feature aggregation is achieved via skip connections. The output of the last transformer block is resized and then concatenated with the previously extracted features to form a set of multi-level features, which will be fed into the invertible network of the conditional normalization flow. During invertible network training, a normal-light image is through 3-level transitions, each comprising a squeeze layer for halving the feature map size, a transition step for pixel rearrangement, and K flow steps for gradually mapping input data to Gaussian distribution. In the Flow step, conditional affine coupling and affine injector are employed to receive features extracted by MSFormer. During inference, reverse the above process, given a low-light image and a Gaussian distribution, the features extracted by MSFormer and the reverse calculation of the invertible network can be used to infer the enhanced image.

3.2. Unbalanced points-guided transformer

In original Transformers, attention is computed across all patches, the long-range attention may come from any image region, regardless of the signal and noise levels. In fact, regions of unbalanced points exhibit significant information distortion and loss. As a result, the

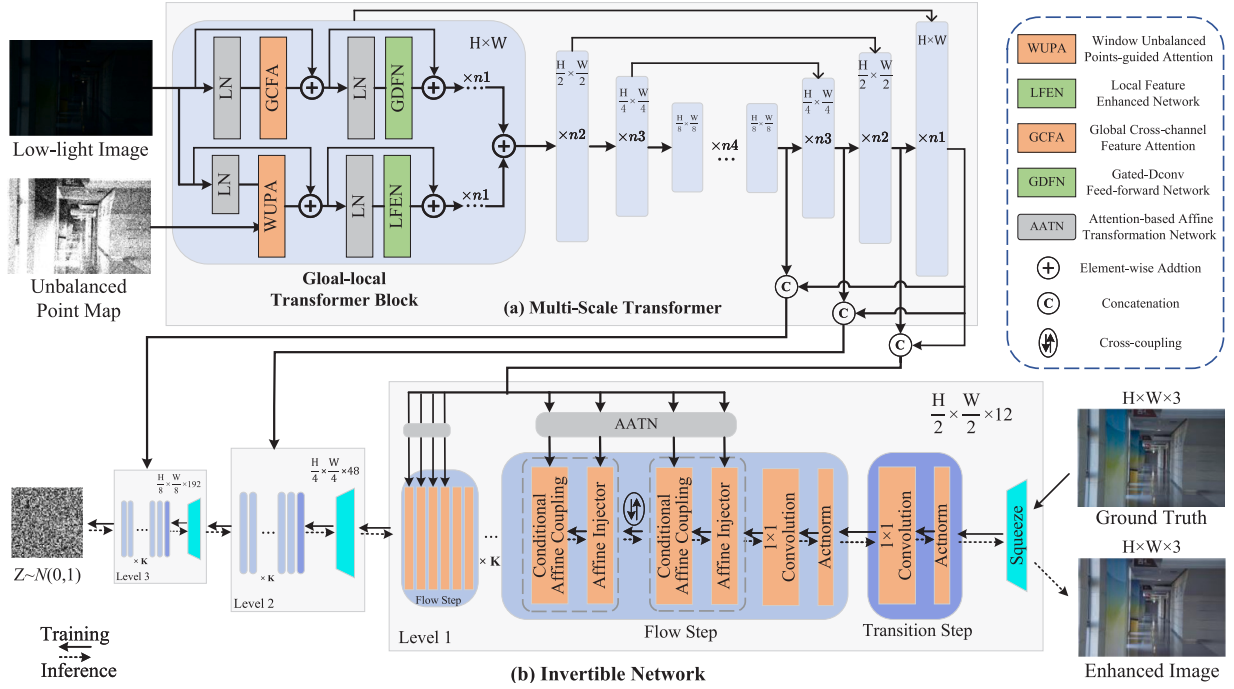


Fig. 4. Overview of our UPT-Flow framework for low-light image enhancement.

information they provide is inaccurate, significantly disrupting image enhancement. To address this, we introduce Window-based Unbalanced Points-guided Attention (WUPA) to improve the Transformer.

Unbalanced point prior. In this paper, Unbalanced points are defined based on the RGB ratio of images. Specifically, for each pixel, the pixel values in the three RGB channels (Red, Green, and Blue) are split, and the ratios of each channel's value to the sum of all channel values are computed. If any of these ratios is less than or equal to a predefined threshold, the pixel is marked as an Unbalanced Point; otherwise, it is marked as a Balanced Point. Mathematically, unbalanced point map M can be represented as:

$$M(x, y) = \begin{cases} 0, & P(x, y) \leq T \\ 1, & P(x, y) > T, \end{cases} \quad (2)$$

where T is the threshold which controls the detection of unbalanced points, (x, y) means the pixel position, and $P(x, y)$ is the minimum RGB ratio in certain pixel (x, y) :

$$P(x, y) = \min(P_R, P_G, P_B) \\ = \min\left(\frac{R(x, y)}{\text{sum}(x, y)}, \frac{G(x, y)}{\text{sum}(x, y)}, \frac{B(x, y)}{\text{sum}(x, y)}\right), \quad (3)$$

where $\text{sum}(x, y)$ is the sum of three channel values:

$$\text{sum}(x, y) = R(x, y) + G(x, y) + B(x, y). \quad (4)$$

Based on formula (2), we constructed a binary map composed of unbalanced points, where pixels labeled as zero exhibit relatively severe information distortion. In attention computation, these points are assigned lower attention weights to avoid the impact of distorted information.

Window unbalanced points-guided attention. The traditional spatial attention [38,39], performs global self-attention over all tokens, resulting in a quadratic computational cost with respect to the amount of tokens. It is not suitable for applying global self-attention on high-resolution feature maps. Therefore, following previous work [40,41], we calculate attention with shifted local windows. These windows are arranged to evenly partition the image in a non-overlapping manner, i.e., the whole input features are divided into $\frac{HW}{M^2}$ non-overlapping

patches of size $\mathbb{R}^{M^2 \times C}$. The formula for standard multi-head self-attention is computed as follows:

$$\text{Atten}(Q, K, V) = V \cdot \text{Softmax}\left(\frac{Q \cdot K^T}{\sqrt{d}} + B\right), \quad (5)$$

where $Q, K, V \in \mathbb{R}^{M^2 \times C}$ are the *query*, *key* and *value* matrices, which can be obtained as tokens from the input feature maps; d is the *query/key* dimension [38–40], which can reduce the range of elements in the matrices to improve the stability of the model, and B is the relative position bias [40,41]. In our unbalanced points-guided attention, V represents tokens obtained from the input feature map, while Q and K are derived from the unbalanced point map M . The attention of WUPA module can be calculated as:

$$\text{Out} = \text{Atten}(M, M, V). \quad (6)$$

Note that, guided by the unbalanced point map prior, those points with severe distortion of information will get lower attention weights, while the areas with better information will have high weights. Regions with severe information distortion mainly contain noisy information, and their information is often inaccurate. If these regions have high weights, they would detrimentally impact image enhancement.

Local feature enhanced network. In previous work [1,39], the Feed-Forward Network (FFN) is based on Multi-Layer Perceptron (MLP), which suffers limited capacity to utilize local context. Actually, neighboring high-quality pixels are crucial for image restoration. Therefore, to better enhance unbalanced regions of images, we design a Local Feature Enhanced Network (LFEN). As illustrated in Fig. 5, we first use a convolutional projection layer on each token to augment its feature dimension. Following that, a 3×3 depth-wise convolution is applied to obtain local information. Subsequently, we reduce the number of channels via an additional convolutional layer to align with the input channel dimension. Finally, we attach an efficient channel attention module [42], which involves only a small number of parameters while delivering significant performance gains.

3.3. Glocal-local transformer block

Although WUPT can extract local features, partitioning windows can lead to constrained self-attention calculations for the windows

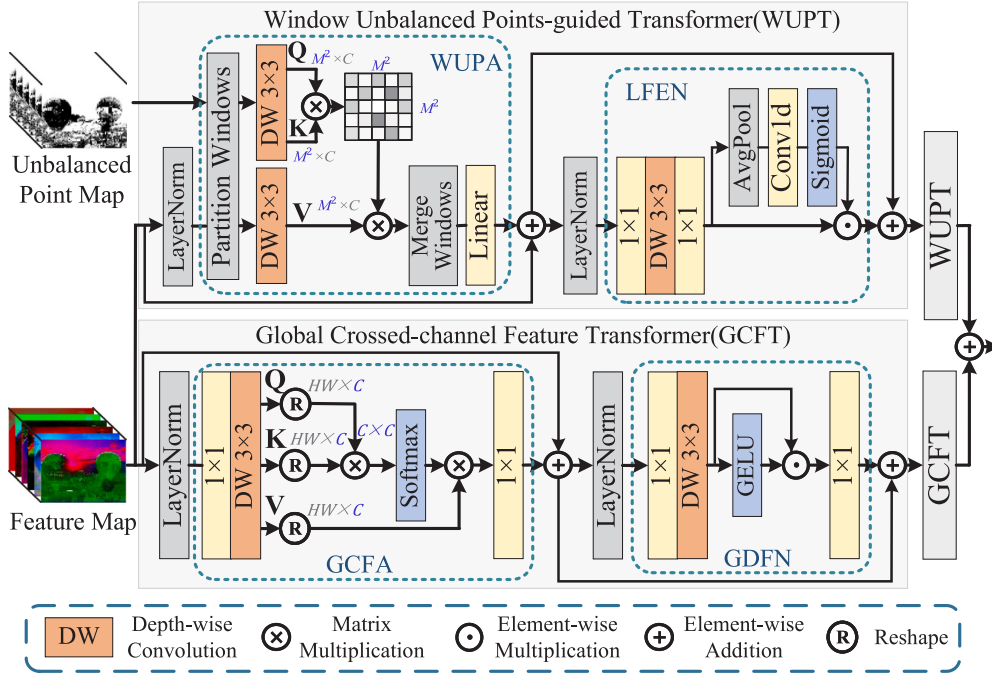


Fig. 5. The network of proposed global-local transformer block, which consists of Window-based Unbalanced Points-guided Transformer blocks (WUPT) and Global Crossed-channel Feature Transformer blocks (GCFT).

positioned at the image boundary, resulting in inadequate interaction between windows. This lack of interaction may result in the loss of crucial global information. Inspired by [43], we propose the use of Global Crossed-channel Feature Transformer (GCFT) to compensate for global information. GCFT computes the self-attention on the crossing channels by the multi-head attention [38] to acquire global context for image enhancement, offering advantages in modeling long-distance dependencies. The Global Crossed-channel Feature Attention (GCFA) can be calculated as:

$$\text{Atten}(Q, K, V) = V \cdot \text{Softmax}(Q \cdot K^T / \alpha), \quad (7)$$

where $Q, K, V \in \mathbb{R}^{HW \times C}$ are derived from a 3×3 depth-wise convolution employed to encode channel-wise spatial context, and α is a trainable scaling parameter [43] to control the magnitude of the dot product of Q and K before applying the softmax function.

Thereby, we construct a global-local transformer block composed of WUPT and GCFT, as shown in Fig. 5. WUPT contributes to the restoration of distorted information in unbalanced point regions, while GCFT focuses on compensating and enhancing global information. The element-wise addition of outputs from WUPT and GCFT achieves efficient information fusion of local and global. By combining several different scale global-local transformer blocks, we construct a U-shaped multi-scale feature fusion module (i.e., MSFormer) for encoding low-light images, which fully considers both global contextual information and local fine-grained details of low-light images. This organic integration of rich and high-quality information aids in conditional normalization flow better modeling of complex image scenes and achieving higher-quality image enhancement results.

Computational complexity. Based on formula (6) and (7), we analyze the attention computational complexity of our proposed global-local transformer block module as:

$$\begin{aligned} O(\text{WUPA}) &= (M^2 HWC), O(\text{GCFA}) = (C^2 HW), \\ O(\text{WUPA} + \text{GCFA}) &= (M^2 HWC + C^2 HW), \end{aligned} \quad (8)$$

where M and C are predefined constants. Thus, the attention calculation of our proposed MSFormer module achieves linear computation costs related to image size.

3.4. Conditional normalizing flow

Normalizing flows can explicitly model the distribution of the target image by optimizing the negative log-likelihood, which can avoid blurring, noise, and artifacts caused by pixel-level reconstruction and adversarial learning. However, flow networks of LLFlow [17] do not fully learn flow features, resulting in its performance lagging behind many SOTA methods. Therefore, we propose two improvements to flow components.

Cross-coupling conditional affine layer. Affine coupling [34,35] is a simple and powerful method to achieve reversible conversion between input and output. However, in previous works [11,17,21], only a fixed half of the flow features perform affine transformations each time, which causes imbalanced learning of flow features. In order to achieve sufficient learning, we propose alternating cross-coupling between each set of affine layers (e.g., h_a^k and h_b^k will take turns performing affine coupling and identity mapping), as shown in Fig. 6(a). A single conditional affine coupling can be expressed as follows:

$$\begin{aligned} h_a^{k+1} &= h_a^k, \\ h_b^{k+1} &= \exp(s_\theta^k(h_a^k; ft)) \cdot h_b^k + t_\theta^k(h_a^k; ft), \end{aligned} \quad (9)$$

where h_a^k and h_b^k are parts of k th layer's h^k in channel dimension. $s_\theta^k(\cdot)$ and $t_\theta^k(\cdot)$ are two arbitrary CNN networks that calculate the scaling and bias of h_b^k . ft is the conditioning variable extracted by MSFormer. Moreover, to further achieve more direct information transfer from low-light images to the flow branch, we use the affine injector layer [11] to apply the affine transformation on the full activations:

$$h^{k+1} = \exp(s_\theta^k(ft)) \cdot h^k + t_\theta^k(ft). \quad (10)$$

We implement Eqs. (9) and (10) together to form our cross-coupling conditional affine layer. Unlike existing flow models that enhance model expressiveness by simply stacking half affine layers, our cross coupling can fully mixing information, further improving the network's non-linearity, and enhancing its fitting capability.

Attention-based affine transformation network. The previous works [11,17,21] only used a 3-layer Neural Network (NN) as $s_\theta^k(\cdot)$ and $t_\theta^k(\cdot)$ in Eqs. (9) and (10) for affine feature transformation, which failed to

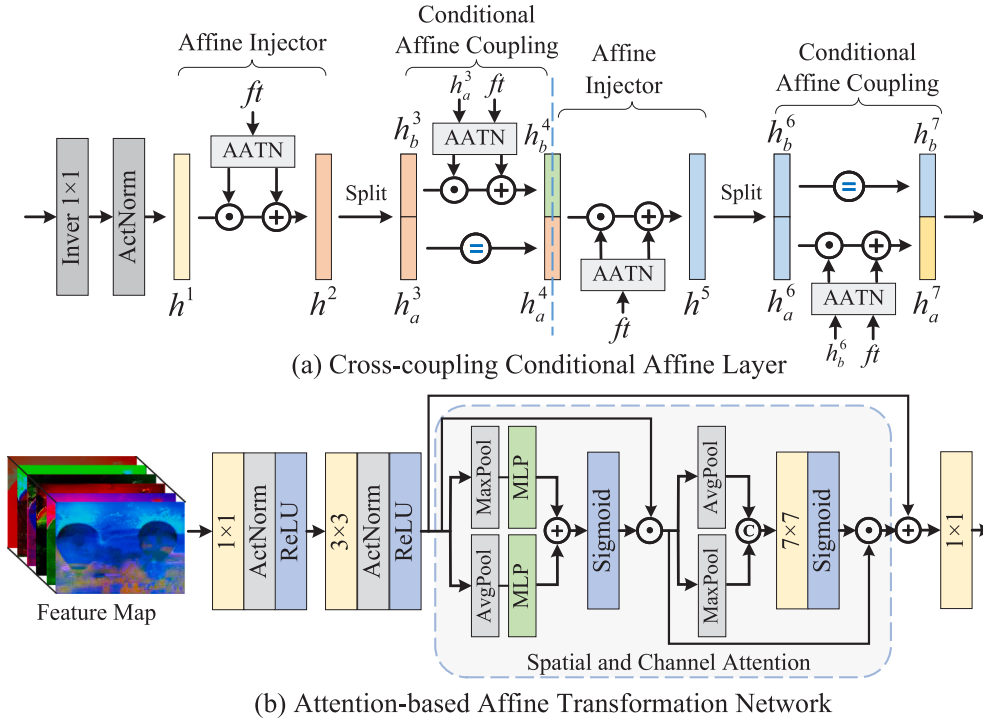


Fig. 6. The components of one flow step. (a) Cross-coupling and identity mapping between each set of conditional affine coupling in one flow step. (b) Spatial and channel attention-based affine transformation network (AATN) in (a).

fully consider the distinct characteristics present in different regions of the image. Based on this, we designed a spatial and channel Attention-based Affine Transformation Network (AATN), as shown in Fig. 6(b). The low-light image condition and normal-light image features are fused by the AATN, which enables the model to dynamically adjust the weights assigned to different parts of the image, accommodating varying degrees of impact from weak illumination. The AATN enhances the expressive ability of a single flow step without a substantial increase in computational overhead and brings a significant boost in performance compared to the baseline.

3.5. Training and inference

Normalizing flows are models for invertible density estimation, aiming to learn a bijective function f_θ mapping a complex data distribution p_Y to a simple predefined prior p_Z . For a sample $\mathbf{y} \in Y$, the probability density $p(\mathbf{y})$ can be explicitly calculated using the change of variable formula:

$$p(\mathbf{y}, \theta) = p_Z(f_\theta(\mathbf{y})) \left| \det \frac{\partial f_\theta}{\partial \mathbf{y}} \right|, \quad (11)$$

where the second component is the volume-scaling determinant of Jacobian $\frac{\partial f_\theta}{\partial \mathbf{y}}$, which enable precise maximum likelihood estimation for $p(\mathbf{y})$. Eq. (11) can be extended to a conditional form to learn the conditional distribution $p(\mathbf{y}|\mathbf{c})$ using normalizing flow given the constraints $\mathbf{c}$:

$$p(\mathbf{y}|\mathbf{c}, \theta) = p_Z(f_\theta(\mathbf{y}; \mathbf{c})) \left| \det \frac{\partial f_\theta}{\partial \mathbf{y}}(\mathbf{y}; \mathbf{c}) \right|. \quad (12)$$

In our work, the conditional feature $\mathbf{c}$ is derived from the input low-light image I_l , while the complex distribution $\mathbf{y}$ is the corresponding ground truth I_g . We construct the conditional normalizing flow network f_θ to directly map $p(I_g|I_l)$ to the Gaussian distribution $p_Z(\mathbf{z})$. Thus, the conditional distribution $p(I_g|I_l, \theta)$ can be implicitly defined through the reverse mapping: $p_Z(\mathbf{z}) \xrightarrow{f_\theta^{-1}} p(I_g|I_l, \theta)$. According to Eq. (12), the probability density $p(I_g|I_l)$ can be calculated as:

$$p(I_g|I_l, \theta) = p_Z(f_\theta(I_g; I_l)) \left| \det \frac{\partial f_\theta}{\partial I_g}(I_g; I_l) \right|. \quad (13)$$

In the training phase, given the condition I_l , the network maps the corresponding normal-light image I_g to the latent variable $\mathbf{z} = f_\theta(I_g; I_l)$. During the inference stage, owing the reversibility of the network, the enhanced image prediction $I_e = f_\theta^{-1}(\mathbf{z}; I_l)$ can be obtained through sampling $\mathbf{z}$ from the Gaussian distribution $\mathbf{z} \sim p_Z(\mathbf{z})$.

The main distinction between normalizing flow and other generative models is that Eq. (13) makes it achievable to compute the probability density of the conditional distribution $p(I_g|I_l, \theta)$ directly. The network can be trained by maximizing $p(I_g|I_l, \theta)$, equivalent to minimizing the negative log-likelihood loss function $\mathcal{L}(\theta; I_g, I_l)$, which can be expressed as follows:

$$\begin{aligned} \mathcal{L}(\theta; I_g, I_l) &= -\log p(I_g|I_l, \theta) \\ &= -\log p_Z(f_\theta(I_g; I_l)) - \log \left| \det \frac{\partial f_\theta(I_g; I_l)}{\partial I_g} \right|. \end{aligned} \quad (14)$$

In fact, the invertible network f_θ comprises several reversible layers, denoted as $f_\theta^0 \circ f_\theta^1 \circ \dots \circ f_\theta^{K-1}$, where $\circ$ represents the nestification of functions. Let h^k and h^{k+1} represent the input and output of each invertible layer f_θ^k , respectively. Then the calculation process of each reversible layer can be formulated as $h^{k+1} = f_\theta^k(h^k, E_\theta(I_l))$, where E is our proposed Multi-Scale Transformer (MSFormer) which encodes low-light images into conditional information $f_t = E_\theta(I_l)$ for our designed cross-coupling conditional affine layers. By expanding f_θ to $f_\theta^0 \circ f_\theta^1 \circ \dots \circ f_\theta^{K-1}$, the negative log-likelihood loss function in Eq. (14) is rewritten as:

$$\begin{aligned} \mathcal{L}(\theta; I_g, I_l) &= -\log p_Z(f_\theta(I_g; I_l)) \\ &\quad - \sum_{k=0}^{K-1} \log \left| \det \frac{\partial f_\theta^k(h^k, E_\theta(I_l))}{\partial h^k} \right|. \end{aligned} \quad (15)$$

The optimization goal of proposed UPT-Flow is given by Eq. (15), and all we need to do is optimize the negative log-likelihood during the training phase to learn the conditional distribution.

3.6. Advantages of maximum likelihood estimation

Likelihood refers to the capability to calculate the probability density or probability value of a given data under the model. The likelihood

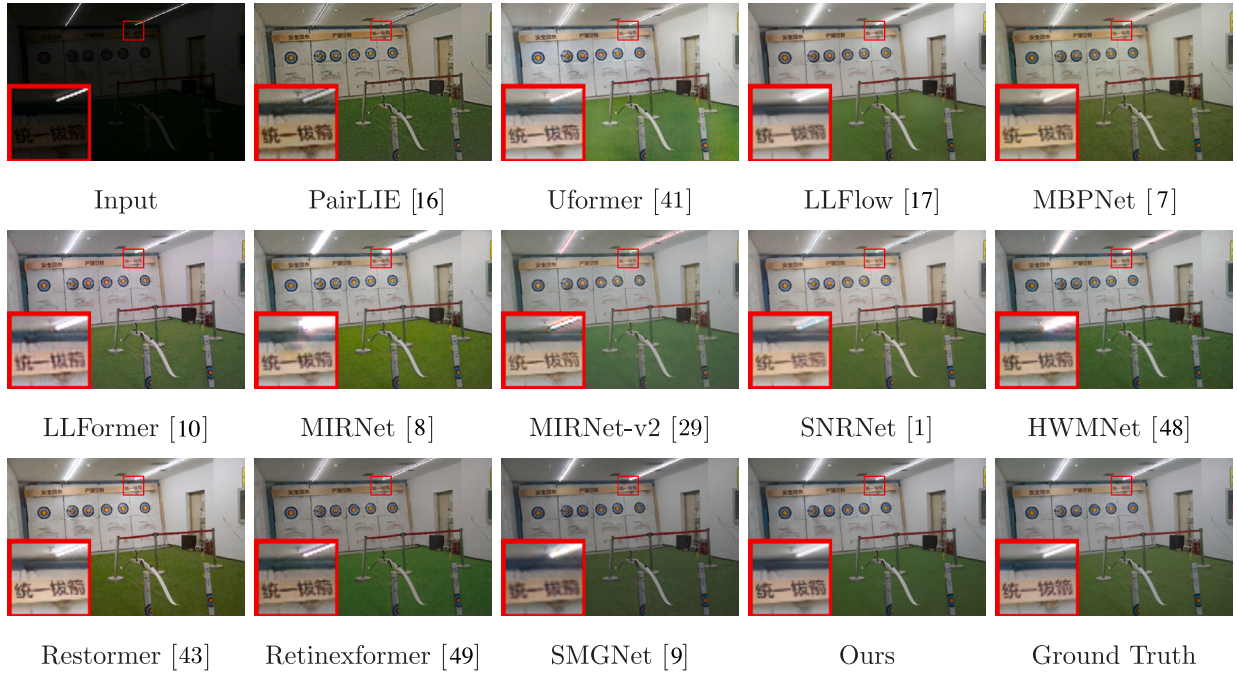


Fig. 7. Visual comparison with state-of-the-art low-light image enhancement methods on LOLv2-real [6].

value reflects how well the model fits the data distribution. In normalizing flows, this is achieved by learning an invertible transformation that maps complex data distributions to simple ones. Because the transformation is both reversible and deterministic, we can accurately compute the log-likelihood of each data point, enabling precise optimization during training.

The advantage of exact likelihood computation lies in its ability to capture fine details in the data distribution, leading to higher fidelity in generated samples. This contrasts with models like GANs and VAEs, which do not compute exact likelihoods. GANs, for example, generate realistic samples through adversarial training but do not explicitly calculate the probability density of these samples, which limits their ability to fully capture the underlying data distribution. VAEs, on the other hand, use variational inference to approximate the posterior distribution, resulting in an approximate likelihood rather than an exact one, which often leads to blurry outputs. Furthermore, models optimized using L1 or L2 loss typically focus on minimizing reconstruction error rather than modeling the data's underlying probability distribution. While this approach is straightforward and effective for certain tasks, it often results in oversmoothing and a loss of fine details in the generated images.

In contrast, maximizing the likelihood in UPT-Flow can obtain the exact likelihood, resulting in a more faithful representation of the data distribution. This enables the model to enhance low-light images with greater sharpness and detail, making it particularly suitable for tasks that require high-quality generation.

4. Experiments

4.1. Datasets and implementation details

We evaluate our method on LOLv2 (real & synthetic) [6], MIT-Adobe FiveK (MIT) [45], SDSD (indoor & outdoor) [44], SMID [46], Night Traffic Monitoring (NTM) [19,23] datasets. These databases have distinct characteristics, enabling a more comprehensive evaluation of methods' performance.

LOLv2. The LOLv2 [6] database is the most authoritative database for low-light image enhancement, which is collected in real scenes and is divided into real and synthetic subsets. The low-light images

in LOLv2 exhibit varying characteristics, with some having extremely low brightness, others showing uneven lighting, and some featuring noticeable noise. The training and testing sets are split in proportion to 689:100 and 900:100 on LOLv2-real and LOLv2-synthetic.

MIT. The MIT [45] dataset encompasses a diverse range of scene types without extremely low brightness. Consistent with prior approaches [7], we utilize the outputs from Expert C. The last 500 paired images are chosen for testing, and trained on the remaining 4500 pairs.

SDSD. The SDSD [44] dataset, like LOLv2, with extremely low brightness but larger image size, which contains indoor and outdoor subsets. Similar to previous methods [1,49], we split the dataset into 1655:307 and 2650:500 for training and testing on SDSD-indoor and SDSD-outdoor, respectively.

SMID. The SMID [46] dataset gathers pairs of RAW images with long and short exposures, comprising a substantial dataset for both training and testing purposes. The RAW data is then converted into pairs of normal and low-light RGB images [1,49]. A total of 15,763 pairs are utilized for training, while 1470 pairs are selected for testing.

NTM. The NTM [19,23] database contains 664 extremely low-brightness RGB images collected by intelligent traffic monitoring systems at night. The NTM images have a lot more noise than expert-captured databases. To better evaluate the performance of the proposed method across different low-light conditions, images in NTM are partitioned into two subsets based on the mean pixel intensity (MPI) value of images. These subsets are named NTM part1 and NTM part2, and their MPI values range from [0, 30] to [30, 50], respectively. NTM part1 contains 582 images, while NTM part2 contains 82 images.

Implementation details. We implement our framework in PyTorch, and all experiments are completed on a PC with an NVIDIA RTX4090 GPU. From $n1$ to $n4$, the number of 4-level symmetric MSformer blocks are [2, 2, 2, 2], and the numbers of flow steps K in the invertible network is set to 8. In formula (2), the threshold T for unbalanced points is set to 0.1. We use Adam as the optimizer with an initial learning rate of 2×10^{-4} and the learning rate decreased with a factor of 0.5 at the maximum iterations of [0.4, 0.6, 0.8, 0.9]. For LOLv2-real and LOLv2-synthetic, we use randomly cropped patch sizes of 160×160 with a window size of 5, whereas for MIT, SMID, and SDSD the patch sizes are 192×192 with a window size of 8. We set the maximum iterations to 50,000, and the whole training process takes approximately 7 h on LOLv2-real.

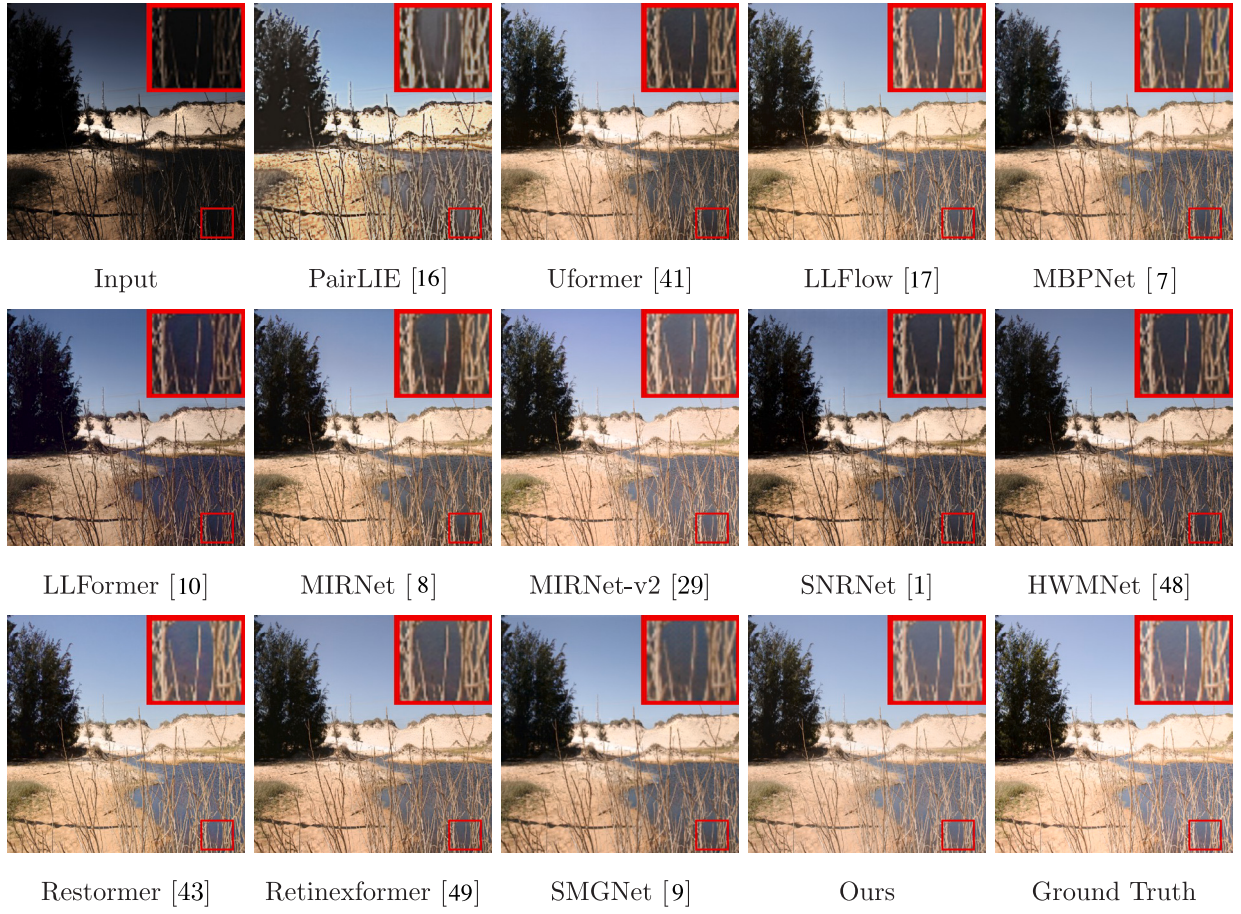


Fig. 8. Visual comparison with state-of-the-art low-light image enhancement methods on LOLv2-synthetic [6].

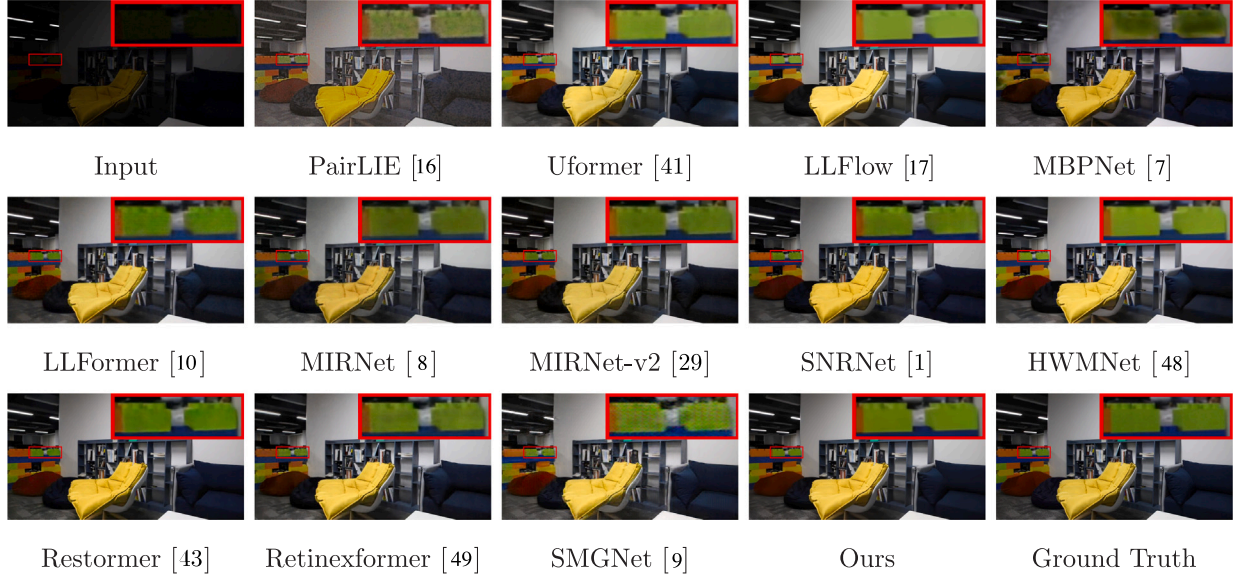


Fig. 9. Visual comparison with state-of-the-art low-light image enhancement methods on SDSD-indoor [44].

4.2. Comparison with current methods

We compare our method with a rich collection of SOTA methods for low-light image enhancement, including the traditional methods: LIME [22]; the zero-shot or unsupervised learning methods: ZeroDCE++ [13], RUAS [14], SCI [15], PairLIE [16], and ZERO-IG [47];

the supervised learning methods: Uformer [41], LLFlow [17], MBPNet [7], LLFormer [17], MIRNet [8], MIRNet-v2 [29], SNRNet [1], Restormer [43], HWMNet [48], Retinexformer [49], and SMGNet [9]. Please note that all these methods' results are reproduced either through using the officially provided pre-trained models or by training with recommended parameters. We adopt the peak signal-to-noise ratio

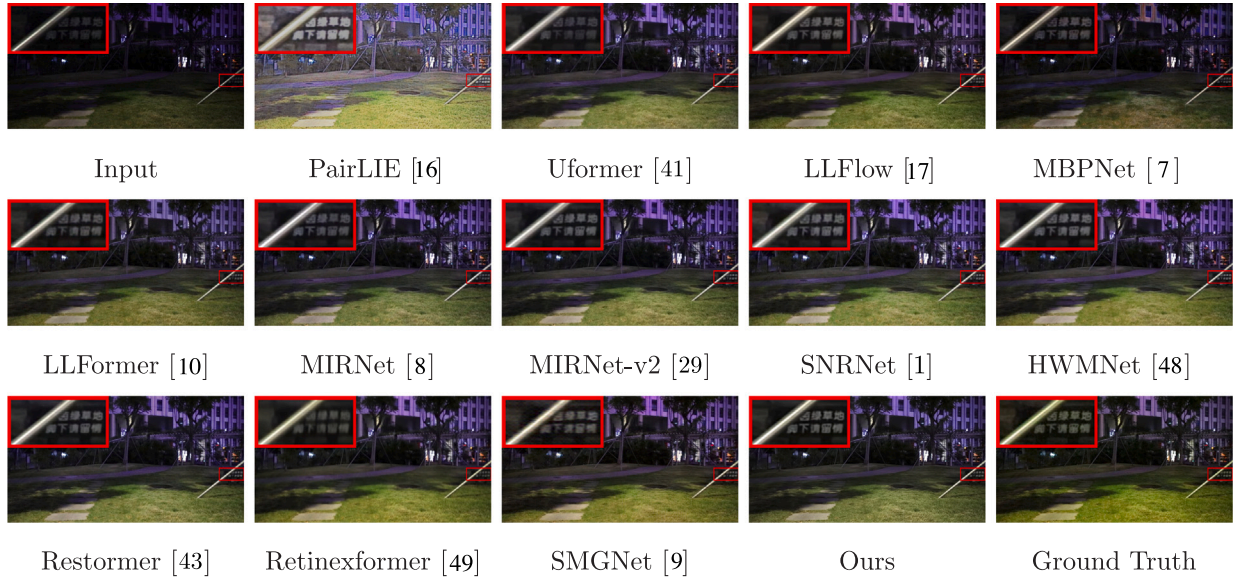


Fig. 10. Visual comparison with state-of-the-art low-light image enhancement methods on SDSD-outdoor [44].

Table 1

Quantitative comparisons on LOLv2-real [6], LOLv2-synthetic [6], MIT [45], and SMID [46]. Our method performs the best consistently.

Methods	Param(M)	LOLv2-real			LOLv2-synthetic			MIT-Adobe FiveK			SMID		
		PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
RUAS [14] CVPR'21	0.0014	15.326	0.493	0.310	13.765	0.634	0.305	17.858	0.813	0.146	18.678	0.639	0.385
SCI [15] CVPR'22	0.0003	17.304	0.540	0.308	15.428	0.744	0.233	20.725	0.845	0.107	19.742	0.653	0.359
LIME [22] TIP'16	/	17.781	0.499	0.353	17.193	0.764	0.200	17.260	0.790	0.112	15.121	0.571	0.464
ZERO-IG [47] CVPR'24	0.12	18.602	0.751	0.388	17.124	0.769	0.215	16.495	0.714	0.289	16.218	0.625	0.475
ZeroDCE++ [13] TPAMI'21	0.79	18.433	0.587	0.306	17.686	0.815	0.186	15.361	0.705	0.225	17.059	0.612	0.405
PairLIE [16] CVPR'23	0.34	19.885	0.773	0.234	19.074	0.794	0.230	13.271	0.691	0.215	13.357	0.532	0.507
Uformer [41] CVPR'22	5.29	17.074	0.805	0.172	23.473	0.916	0.077	22.266	0.852	0.073	26.236	0.785	0.199
LLFlow [17] AAAI'22	37.68	19.670	0.852	0.157	22.376	0.910	0.066	24.695	0.925	0.450	28.117	0.813	0.181
MBPNet [7] TIP'23	16.96	19.948	0.837	0.144	24.546	0.918	0.066	25.318	0.923	0.043	29.632	0.771	0.177
LLFormer [10] AAAI'23	24.52	20.990	0.811	0.219	23.741	0.902	0.086	25.295	0.922	0.049	27.917	0.785	0.183
MIRNet [8] ECCV'20	31.76	21.180	0.840	0.145	25.083	0.920	0.070	24.022	0.901	0.059	28.617	0.810	0.180
MIRNet-v2 [29] TPAMI'22	5.86	21.372	0.833	0.153	24.292	0.923	0.064	24.593	0.900	0.062	28.488	0.804	0.174
SNRNet [1] CVPR'22	39.12	21.479	0.849	0.157	24.139	0.928	0.056	24.902	0.901	0.101	28.493	0.805	0.178
Restormer [43] CVPR'22	26.12	21.540	0.841	0.154	25.476	0.934	0.051	25.236	0.888	0.055	28.181	0.800	0.173
HWMNet [48] ICIP'22	66.56	21.713	0.846	0.157	25.012	0.927	0.056	24.441	0.914	0.044	28.971	0.817	0.172
Retinexformer [49] ICCV'23	1.61	22.794	0.839	0.171	25.670	0.928	0.059	24.556	0.883	0.069	28.990	0.811	0.167
SMGNet [9] CVPR'23	18.36	24.032	0.820	0.169	24.980	0.894	0.092	25.227	0.854	0.047	26.971	0.725	0.212
Ours	23.45	25.056	0.890	0.115	26.481	0.946	0.037	25.667	0.928	0.041	29.529	0.822	0.167

(PSNR), structural similarity (SSIM) [50] (non-grayscale), and learned perceptual image patch similarity (LPIPS) [51] as the evaluation metrics, and without using any tricks.

Qualitative analysis. The visual comparisons of our method and other SOTA algorithms are shown in Figs. 7, 8, 9 and 10. Previous models trained with L1 loss tend to have severe detail blurring, such as LLFormer, SNRNet, Retinexformer, and SMGNet in Figs. 7 and 10. Moreover, there are overexposure and color distortion in MIRNet-v2, HWMNet, and Uformer in Fig. 7. Additionally, there are imbalanced and insufficient enhancement of different regions of images, as shown in Fig. 8, as well as obvious visual artifacts, as shown in Fig. 9. In contrast, our method shows better visual quality, more accurate and clear details, more natural color consistency, and accurate brightness, achieving the closest perceptual similarity to ground truth.

Quantitative results. The objective evaluation metrics are shown in Tables 1 and 2, where the best scores are **highlighted** in bold. Overall, our method significantly outperforms SOTA methods across all databases and substantially improves the best metrics without excessive memory overhead. We achieve the highest PSNR and SSIM, meaning that our method provides the most suitable brightness and contrast, the lowest distortion, and more high-frequency details and structures.

Additionally, lower LPIPS indicate a closer alignment with human perception in terms of color and texture.

4.3. Night traffic monitoring enhancement

Images captured under night traffic monitoring do not have corresponding normal-light images as references. Therefore, we employ three no-reference image quality assessment metrics, LOE, NIQE, and BRISQUE, as key criteria to measure the quality of enhanced images. LOE is based on brightness, representing the lightness order error that reflects the naturalness of an enhanced image. NIQE and BRISQUE are both evaluation models constructed based on statistical features of natural scene images (including image gradient, contrast, brightness, etc.). The lower the NIQE and BRISQUE, the less distorted and more natural the image is. For LOE, the smaller the LOE value is, the better the lightness order is preserved.

Quantitative results. Table 3 presents the objective evaluation results of the proposed UPT-Flow on two different brightness subsets of NTM. Overall, our method significantly outperforms various SOTA methods. Through a more detailed analysis of indicators, it can be found that among the top 5 methods with the best overall performance, 4 are

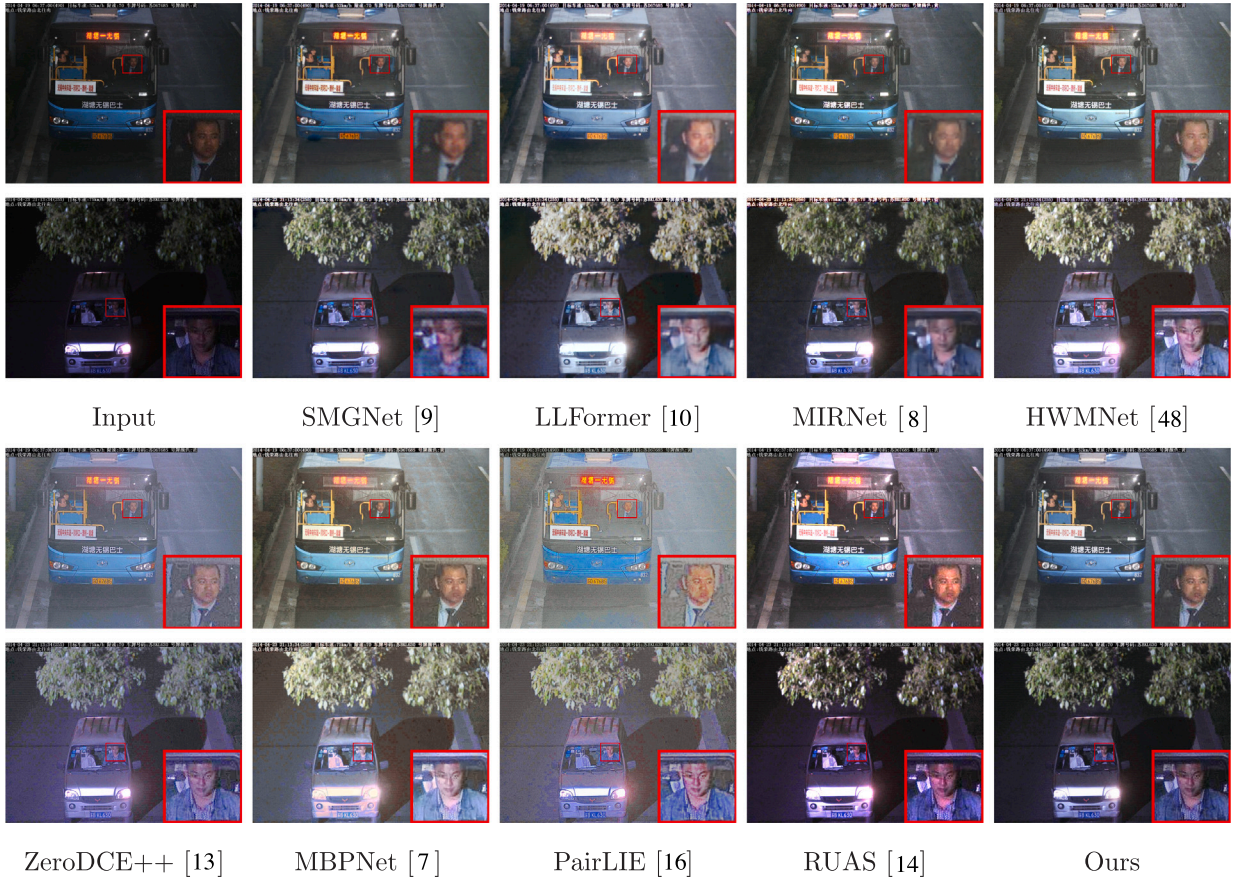


Fig. 11. Visual comparisons on NTM [19,23]. “Ours” has less noise, more natural colors and more moderate brightness.

Table 2

Quantitative results on SDSD-indoor [44] and SDSD-outdoor [44]. Our method significantly outperforms SOTA algorithms.

Methods	SDSD-indoor			SDSD-outdoor		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$
RUAS [14]	16.266	0.651	0.345	20.847	0.689	0.234
SCI [15]	21.194	0.723	0.328	18.095	0.538	0.296
LIME [22]	20.435	0.569	0.410	12.808	0.381	0.398
ZERO-IG [47]	24.196	0.815	0.255	10.334	0.423	0.533
ZeroDCE++ [13]	20.530	0.671	0.360	16.074	0.484	0.352
PairLIE [16]	19.460	0.689	0.284	12.368	0.434	0.368
Uformer [41]	25.471	0.876	0.130	26.771	0.841	0.160
LLFlow [17]	25.459	0.896	0.139	28.823	0.869	0.188
MBPNet [7]	28.822	0.882	0.126	22.872	0.754	0.186
LLFormer [10]	29.650	0.874	0.152	28.734	0.838	0.129
MIRNet [8]	27.832	0.882	0.138	29.167	0.871	0.152
MIRNet-v2 [29]	28.738	0.890	0.122	30.067	0.866	0.152
SNRNet [1]	29.440	0.894	0.129	28.660	0.866	0.140
Restormer [43]	30.026	0.901	0.114	27.987	0.863	0.166
HWMNet [48]	29.817	0.905	0.119	27.937	0.870	0.158
Retinexformer [49]	29.749	0.892	0.118	29.833	0.875	0.178
SMGNet [9]	26.889	0.802	0.166	26.329	0.809	0.176
Ours	31.197	0.912	0.110	30.214	0.887	0.174

unsupervised methods, meaning that unsupervised methods tend to show more robust performance when confronted with real-world open images. On the contrary, our method not only demonstrates a significant advantage over a host of supervised methods but also shows improvement over the latest unsupervised methods. To be specific, LOE is 22.78 and 11.08 better than the best unsupervised method RUAS, and BRISQUE is 3.91 and 1.02 better than the best supervised method MBPNet. This indicates that the enhancement of real nighttime traffic surveillance images using the proposed method in this paper achieves

Table 3

Quantitative results on NTM [19,23]. “BRIS” represents the BRISQUE metric.

Methods	NTM part1			NTM part2		
	LOE $\downarrow$	NIQE $\downarrow$	BRIS $\downarrow$	LOE $\downarrow$	NIQE $\downarrow$	BRIS $\downarrow$
RetinexNet [25]	669.22	8.07	41.58	971.10	6.84	36.53
LIME [22]	623.11	6.68	42.22	445.46	5.39	32.88
SMGNet [9]	409.05	5.14	47.53	310.19	5.01	44.87
LLFormer [10]	406.24	4.70	42.41	299.19	4.49	39.46
MIRNet [8]	370.45	7.24	51.84	283.08	7.47	52.61
SNRNet [1]	328.30	4.97	40.21	194.14	4.82	31.11
ZeroDCE [12]	322.99	5.23	41.25	317.34	4.65	38.36
KinD++ [28]	289.27	6.08	40.25	1093.75	5.80	36.75
HWMNet [48]	284.49	4.47	39.22	182.03	4.51	29.26
ZeroDCE++ [13]	234.83	6.20	41.71	278.42	5.13	34.12
MBPNet [7]	231.38	5.31	39.79	197.78	4.73	29.57
PairLIE [16]	213.76	5.03	39.38	237.80	5.03	36.01
RUAS [14]	188.52	5.86	36.52	105.54	5.23	34.57
Ours	165.75	5.38	35.88	94.46	4.47	28.55

superior brightness and more natural enhancement results in objective evaluations. It can be seen that our proposed method can achieve superior brightness and more natural results in night traffic monitoring enhancement.

Qualitative analysis. The visual comparison of NTM is shown in Fig. 11. Low-light images from NTM have extremely low brightness and severe information loss, thus posing challenges to most methods for enhancement. The three supervised methods SMGNet, LLFormer, and MIRNet cannot achieve satisfactory results, whose enhanced images have significant blurring, particularly the driver faces. The enhanced images of MBPNet and PairLIE exhibit severe noise and color distortion. ZeroDCE++ can effectively suppress overexposure, but it also brings strong noise and lower saturation of the enhanced images. The

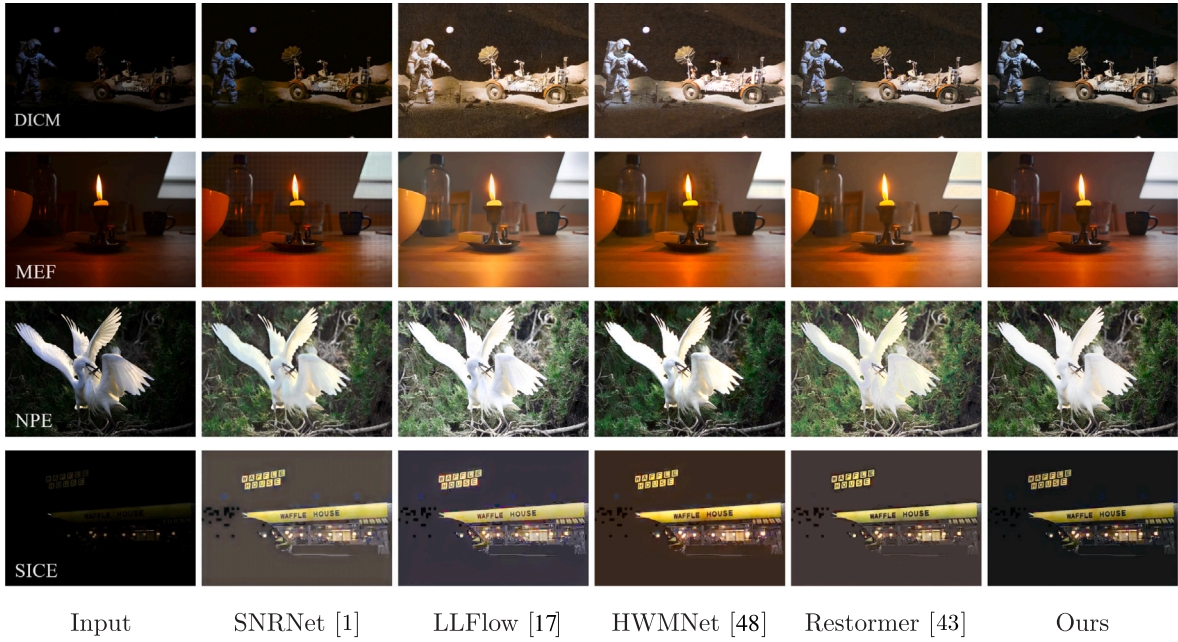


Fig. 12. Visual comparisons on unpaired datasets. “Ours” have less noise and clearer visibility.

Table 4

User study scores on five unpaired datasets.

Methods	DICM	LIME	MEF	NPE	SICE	Mean $\uparrow$
PairLIE [16]	2.78	2.58	2.60	2.56	2.92	2.69
LLFlow [17]	2.26	3.23	3.57	2.84	3.06	2.99
MBPNet [7]	2.10	2.81	3.06	2.66	2.93	2.71
SNRNet [1]	2.66	2.67	2.60	3.04	3.07	2.80
Restormer [43]	3.22	3.47	3.54	3.41	3.17	3.36
HWMNet [48]	2.94	3.33	3.31	3.37	3.18	3.22
SMGNet [9]	2.16	3.13	2.93	2.82	2.84	2.78
Ours	3.57	3.73	3.83	3.76	3.57	3.69

enhanced images of RUAS have severe color over-enhancement and overexposure problems in the vehicle body and driver’s facial area. In contrast, our method has the most moderate brightness, the lowest noise, the most natural colors, and the clearest detail structure, which also aligns with the results of the three objective evaluation metrics.

4.4. User study

To further evaluate the model’s robustness and generalization, we also test our method on five unpaired datasets: DICM [52], LIME [22], MEF [53], NPE [54], and SICE [55] which have no ground truth. In addition, to better quantify whether the enhanced images align with human perception, we employed user study scores and visual comparison to quantify the human subjective perceptual quality of enhanced low-light images.

We conducted a large-scale user study involving 142 participants who were tasked with independently scoring the visual quality of enhanced results. Testers were instructed to assess the results based on four criteria: (1) presence of under/over-exposed regions, (2) occurrence of color distortion, (3) clarity of details, and (4) presence of noise or artifacts. Scores ranged from 1 (worst) to 5 (best). Each low-light image, along with its enhancements from various algorithms (without their names), was presented to testers. A total of 96 testing images were included in the study. The user study scores are presented in Table 4. Our method achieved the highest score compared to other

state-of-the-art supervised and unsupervised algorithms, indicating that our enhanced results were preferred by human subjects.

In addition, some visual comparison results are shown in Fig. 12. As can be seen, our method achieves the most natural enhancement results across various scenes.

4.5. Unbalanced point analysis

4.5.1. Unbalanced point threshold selection

The threshold T for unbalanced points determines which points in the low-light image are classified as unbalanced points. This threshold is a predefined value based on the image characteristics. We conducted extensive experiments on the LOLv2-real and SDSD-indoor datasets to analyze the impact of the unbalanced point threshold T , with the results presented in Fig. 13.

As shown in Fig. 13, when the unbalanced point threshold T is low (less than 0.1), fewer points are marked as unbalanced points and cannot cover enough information distortion regions, contributing in insufficient performance of the proposed model. As T value increases, more points with severe information distortion are marked as unbalanced points, leading to an improvement in model performance. However, when the unbalanced point threshold T becomes too high (greater than 0.1), points with less distortion or points without distortion are also classified as unbalanced points. Due to the fact that unbalanced regions are assigned lower weights in the attention computation of WUPT, the model may not capture adequate and high-quality features from these less distorted regions, leading a significant decline in model performance (as discussed in Section 4.9.2).

The unbalanced point threshold T is a hyperparameter that can be tuned according to different image characteristics. Adjusting threshold T might yield better performance across various datasets. However, in this paper, we do not perform fine-tuning of this parameter. Instead, we only fix it to 0.1 based on experimental experience to demonstrate the effectiveness of our proposed unbalanced point prior and the broad applicability of UPT-Flow.

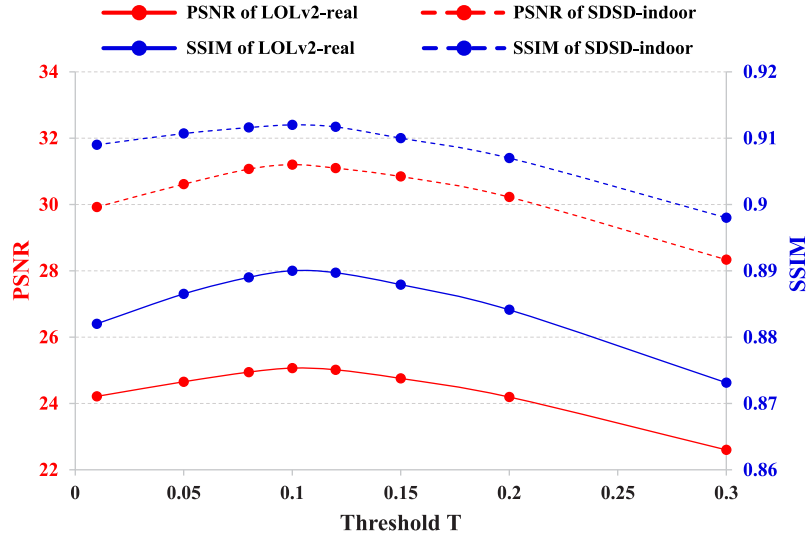


Fig. 13. Performance curves of different unbalanced point thresholds T on LOLv2-real [6] and SDSD-indoor [44] datasets. The solid lines represent LOLv2-real, while the dashed lines represent SDSD-indoor. Red indicates PSNR, and blue indicates SSIM. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

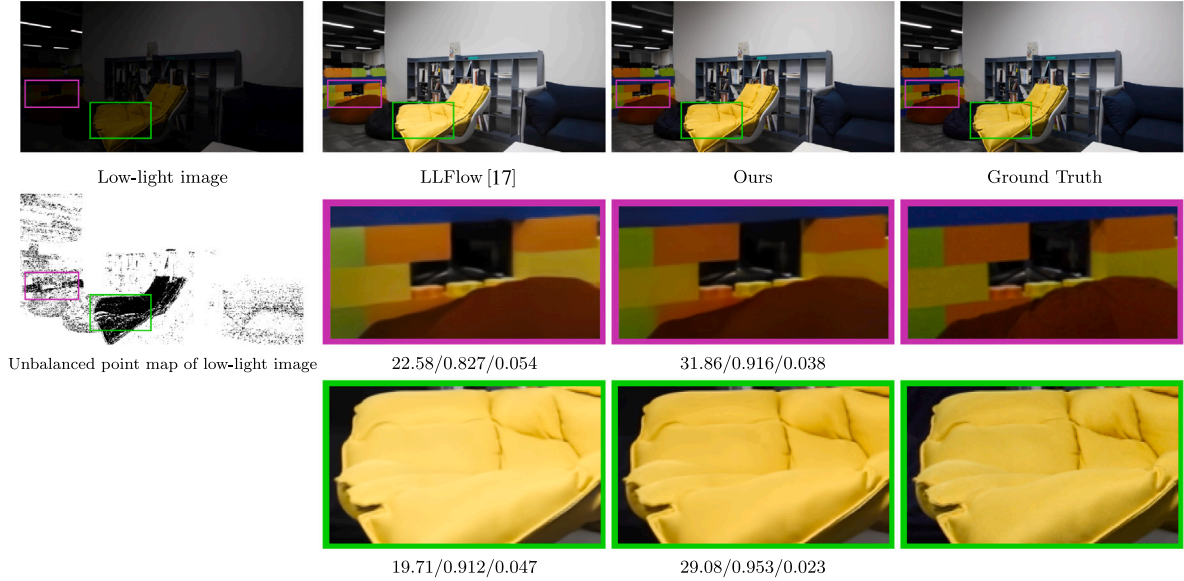


Fig. 14. Visualization of the unbalanced point map and enhanced low-light image. The unbalanced point regions of the enhanced image is magnified and visualized in the corresponding last two rows, and the regional PSNR, SSIM, and LPIPS scores are listed below the enlarged image in order.

4.5.2. Unbalanced points regions analysis

In this section, we further analyze the regions with extremely low brightness and severe information distortion in the images.

In Fig. 14, we visualize the unbalanced point map of the low-light images in the SDSD-indoor [44], which intuitively presents the distribution of unbalanced points. We also visualize the enhancement results of LLFlow and our method, with a magnified view of the unbalanced point regions. There are a large number of unbalanced points in the purple box of the low-light image, and LLFlow fails to correct these distorted information, resulting in color distortion with Ground Truth, as also evidenced by the three metrics listed below the image. Similarly, in the last row of Fig. 14, it can be seen that LLFlow fails to accurately correct brightness, resulting in excessively high brightness. In contrast, the enhancement results of our method, especially in the regions with information distortion, not only achieve the most natural visual outcomes, but also obtain the best results on objective evaluation metrics.

In summary, our proposed unbalanced point prior can accurately restore regions with severe information distortion and achieve desirable results.

4.6. Ablation study

We conduct comprehensive ablation studies to assess the effectiveness of each component in our proposed UPT-Flow. Table 5 and Fig. 15 present the quantitative results and visual comparisons, respectively.

In Table 5, (a) is a baseline constructed using components from the previous normalizing flow. (b) utilizing only GCFT for global information extraction results in slightly higher performance than the baseline. (c) and (d) using our designed LFEN achieve significantly improved performance compared to the previous feed-forward network, and our proposed WUPT also outperforms GCFT, once again proving the necessity of correcting distorted information in unbalanced point regions. (e) including all components of MSFormer, which fully considers global

Table 5

Ablation study of each components of proposed method on LOLv2-real, LOLv2-synthetic, and SDSD-indoor.

	MSFormer					Transformation network		Conditional affine layer		LOLv2-real		LOLv2-synthetic		SDSD-indoor	
	RRDB ^a	GCFT	WUPA	FFN ^b	LFEN	NN ^a	AATN	Simple stacking ^a	Cross-coupling	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
(a)	✓					✓		✓		20.671	0.857	23.432	0.917	26.018	0.897
(b)		✓				✓		✓		21.584	0.862	23.994	0.921	26.989	0.899
(c)			✓	✓		✓		✓		22.372	0.868	24.571	0.926	27.680	0.902
(d)			✓		✓	✓		✓		23.284	0.872	25.021	0.932	28.593	0.904
(e)		✓	✓		✓	✓		✓		24.043	0.881	25.764	0.939	29.741	0.908
(f)		✓	✓		✓		✓	✓		24.824	0.886	26.223	0.943	30.687	0.910
(g)		✓	✓		✓		✓		✓	25.056	0.890	26.482	0.946	31.197	0.912

^a RRDB [56], NN, and Simple stacking are standard components used in previous work [11,17,21].^b FFN is a feed-forward network in previous work [1,38,40].**Fig. 15.** Visual comparison of ablation experiments for checking the effectiveness of the proposed modules. The PSNR, SSIM, and LPIPS scores are listed below the image in order.**Table 6**

Quantitative comparisons of computational complexity in terms of runtime (in second), and FLOPs (in G).

Methods	FLOPs(G)	Inference times(s)
SNRNet [1]	95.83	0.021
HWMNet [48]	452.96	0.079
MIRNet-v2 [29]	563.67	0.022
Restormer [43]	563.96	0.034
LLFlow [17]	1145.35	0.147
MIRNet [8]	3264.62	0.168
Ours	565.38	0.459

context and local detail information, further enhancing performance. (f) and (g) use our designed AATN module in the invertible network of flow and employ cross-coupling to effectively blend features, enhancing the non-linear capability of flow, and ultimately achieving the best performance.

In Fig. 15, we present the visual comparisons on LOLv2-real based on the schemes (a)–(g) from Table 5, and list the objective evaluation metrics above the images. (a) and (b) do not use the proposed unbalanced points-guided attention, resulting in excessive brightness and color distortion. After applying the window-based unbalanced points-guided attention, the overall brightness and color of the image are corrected, but due to the limitations of its local characteristic, the image exhibits lower brightness and blurred details, as shown in Fig. 15(c). When both GCFT and WUPA are used for global and local feature extraction and fusion, the image achieved brightness and color closest to the ground truth, as shown in Fig. 15(e). AATN and alternating cross-coupling significantly enhance the expression ability of the normalizing flow, obtaining the clearest detail structures, as shown in Fig. 15(f) and (g).

4.7. Computational complexity analyses

In Table 1, we demonstrate the parameters of various methods, showing that our proposed model achieves SOTA results with relatively low parameter complexity. In this section, we analyze the computational complexity of our method, as presented in Table 6. Specifically, we calculated the FLOPs for a 512×512 tensor on an RTX 4090 GPU, and inferred 100 RGB images of 512×512 size to compute the average inference time of the models.

As shown in Table 6, the MIRNet has quite large FLOPs, which can be attributed to the recursive residual and the multi-scale feature dense fusion design, leading to substantial computational resource requirements. The FLOPs of our model are comparable to those of MIRNet-v2 and Restormer, which is at a moderate level. More importantly, the FLOPs of our model are only half that of LLFlow, primarily because the RRDB module in LLFlow contains numerous dense blocks and residual connections, resulting in a higher computational overhead due to excessive redundant connections. Our proposed MSFormer avoids redundant connections, optimizing the network structure to significantly reduce computational complexity.

However, our proposed method exhibits the longest inference time in terms of forward propagation. This is mainly due to the extensive use of Depth-wise Convolution in our network. As analyzed in ShuffleNet-v2 [57], Depth-wise Convolution reduce the number of parameters and computational complexity, but it significantly increase the memory access cost (MAC). Specifically, Depth-wise Convolution greatly increases the amount of data access, thereby reaching the memory bandwidth between the GPU and the memory. The I/O limitation replaces the GPU's computing resources as the bottleneck that restricts the model's inference speed.

Table 7
Quantitative results of low-light image object detection on ExDark.

Methods (AP)	Bicycle	Boat	Bottle	Bus	Car	Cat	Chair	Cup	Dog	Motor	People	Table	Mean $\uparrow$
Baseline	74.1	63.4	63.0	83.5	76.4	55.9	51.8	59.1	66.3	61.1	68.4	43.8	63.9
RetinexNet [25]	73.8	62.8	64.8	84.9	80.8	53.4	57.2	68.3	61.5	51.3	65.9	43.1	64.0
RUAS [14]	72.0	62.2	65.2	72.9	78.1	57.3	62.4	61.8	60.2	61.5	69.4	46.8	64.2
KinD [27]	72.2	66.5	58.9	83.7	74.5	55.4	61.7	61.3	63.8	63.0	70.5	47.8	65.0
ZeroDCE [12]	75.8	66.5	65.6	84.9	77.2	56.3	53.8	59.0	63.5	64.0	68.3	46.3	65.1
SCI [15]	74.6	65.3	65.8	85.4	76.3	59.4	57.1	60.5	65.6	63.9	69.1	45.9	65.6
MIRNet [8]	71.8	63.8	62.9	81.4	71.1	58.8	58.9	61.3	63.1	52.0	68.8	45.5	63.6
Restormer [43]	76.2	65.1	64.2	84.0	76.3	59.2	53.0	58.7	66.1	62.9	68.6	45.0	64.9
SNRNet [1]	75.3	64.4	63.6	85.3	77.5	59.1	54.1	59.6	66.3	65.2	69.1	44.6	65.3
Retinexformer [49]	76.3	66.7	65.9	84.7	77.6	61.2	53.5	60.7	67.5	63.4	69.5	46.0	66.1
Ours	82.2	77.8	77.4	91.7	84.0	72.3	70.6	77.7	79.3	78.6	82.3	55.5	77.4



Fig. 16. Visual results of object detection in low-light images (top) and enhanced images (bottom) by our method on ExDark [2].

4.8. Downstream tasks

4.8.1. Low-light object detection

The enhanced images are expected to have better downstream performance than the original ones. We first verify this on low-light object detection.

Experiment settings. Following [58,59], We conduct low-light object detection experiments on the ExDark [2] dataset to assess the preprocessing impacts of various enhancement methods for high-level vision understanding. The ExDark dataset comprises 7363 real-world low-light RGB images, spanning from twilight to extremely dark environments and featuring 12 object classes: Bicycle, Boat, Bottle, Bus, Car, Cat, Chair, Cup, Dog, Motorbike, People, and Table. ExDark is extensively used for low-light object detection due to its 12 object category bounding boxes. For our experiments, we used 80% of the images from each category for training and reserved the remaining 20% for testing. YOLO-v3 [60] served as the detector and was trained from scratch. Different low-light enhancement methods were employed as preprocessing modules, each with fixed parameters.

Quantitative results. The average precision (AP) scores are presented in Table 7. The higher AP scores obtained with our method demonstrate a significant advantage over the previous methods. Specifically, the mean AP of our method is 11.8 AP points higher than the recent best unsupervised method SCI [15] and surpasses that of the second-best fully-supervised method, Retinexformer [49], by an astonishing margin of 11.3 AP points. It is noteworthy that our method achieves the best results across all 12 object categories, providing compelling evidence of its efficacy in enhancing high-level vision understanding.

Qualitative analysis. Fig. 16 depicts a visual comparison of detection results in the low-light (top) scene and the scene enhanced (bottom) by our method. In the first line, the detector often misses recognition or

makes recognition classification errors in underexposed images, as seen in (a) where a desk lamp is mistakenly identified as “people”, and in (b) where a table and multiple cups are missed recognition. Especially in extremely dark environments, detectors often struggle to effectively recognize objects, such as in (c) where ships are obscured by darkness in the river course. In contrast, images enhanced by our method allow the detector to reliably predict well-positioned bounding boxes to cover all objects on the enhanced images. This demonstrates the effectiveness of our method in favoring high-level visual perception.

4.8.2. Nighttime semantic segmentation

Lastly, we extend our experimentation to image semantic segmentation tasks.

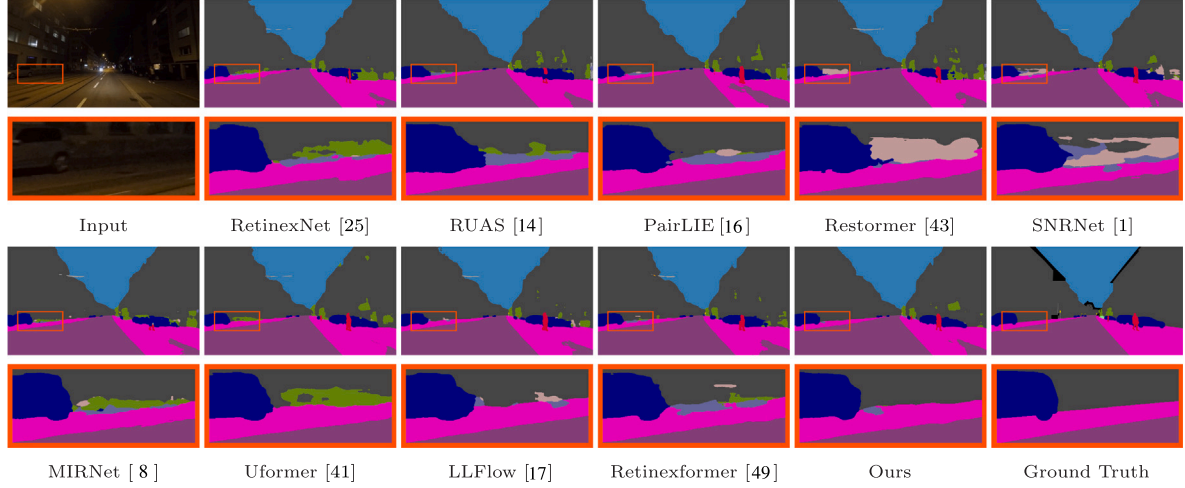
Experiment settings. We conduct low-light image semantic segmentation experiments on the ACDC [3] database. The ACDC [3] database contains a huge collection of 4006 pictures for image semantic segmentation, divided over four frequently unfavorable conditions: rain, night, snow, and fog. Each adverse-condition image includes a high-quality fine pixel-level semantic annotation, a comparable image of the same scene obtained under normal settings, and a binary mask that distinguishes between intra-image regions with clear and ambiguous semantic content. In this experiment, we only use 506 images collected under nighttime conditions, of which 400 are trained and 106 are tested. We adopt DeepLab-V3+ [61] to train from scratch, trained on the ACDC-night train set, and tested on the ACDC-night val set. Different low-light enhancement methods serve as the preprocessing modules with fixed parameters.

Quantitative results. We employ intersection-over-union (IoU) as the metric to evaluate segmentation performance, as shown in Table 8. Our method achieves the highest result on average, 60.4 mIoU, which is 2.2 mIoU higher than the recent supervised image restoration method

Table 8

Quantitative results of nighttime semantic segmentation on the ACDC. The symbol set {RO, SI, BU, WA, FE, PO, TL, TS, VE, TE, SK, PE, CA, TR, BI} represents {road, sidewalk, building, wall, fence, pole, traffic light, traffic sign, vegetation, terrain, sky, person, car, train, bicycle}.

Methods (IoU)	RO	SI	BU	WA	FE	PO	TL	TS	VE	TE	SK	PE	CA	TR	BI	Mean $\uparrow$
Baseline	92.6	71.6	77.0	36.0	31.0	53.4	55.5	25.0	67.3	11.9	79.3	34.0	64.1	81.7	47.0	55.2
RetinexNet [25]	92.9	71.7	75.8	41.9	37.4	52.7	59.0	39.3	67.6	10.0	80.1	36.6	62.6	76.7	28.9	55.6
RUAS [14]	92.3	70.1	76.8	37.4	30.8	50.2	49.0	39.4	68.1	13.3	81.5	26.4	56.9	80.6	45.8	54.6
PairLIE [16]	92.7	70.9	77.2	39.6	37.7	54.4	60.1	43.0	69.0	12.3	80.9	28.3	65.3	86.3	45.6	57.4
Restormer [43]	92.6	70.8	76.3	42.1	37.0	50.5	43.4	40.2	68.1	10.4	79.6	33.6	62.5	76.5	43.2	55.1
SNRNet [1]	93.1	71.2	77.2	42.1	40.1	50.4	52.9	45.3	67.3	11.1	78.6	34.5	62.7	84.7	42.9	57.0
MIRNet [8]	93.0	72.6	78.5	44.9	42.8	51.3	55.3	46.4	69.8	9.5	82.5	29.5	37.6	81.2	47.7	58.2
Uformer [41]	93.8	74.1	77.2	40.0	39.3	53.2	56.0	43.5	68.3	12.0	81.4	34.9	51.0	81.7	48.1	57.2
LLFlow [17]	93.0	73.7	78.1	45.0	42.3	53.5	57.3	43.3	69.3	11.9	82.1	27.5	47.1	84.9	49.8	57.3
Retinexformer [49]	92.4	71.6	76.2	44.2	34.2	51.7	56.8	41.7	68.3	9.8	80.4	29.4	61.8	85.0	42.7	56.4
Ours	93.8	74.2	78.9	45.0	38.0	54.5	55.9	46.9	70.6	12.0	82.5	43.3	73.2	86.9	50.2	60.4

**Fig. 17.** Visual comparison of semantic segmentation on ACDC [3].

MIRNet [8], and 3.0 mIoU higher than the latest unsupervised method PairLIE [16]. Besides, our method yields the best results on 12 object categories: road, sidewalk, building, wall, pole, traffic sign, vegetation, sky, person, car, train, and bicycle.

Qualitative analysis. Fig. 17 shows the semantic segmentation results on the ACDC-night val set. As shown in the zoomed-in regions, except for our method and LLFlow [17], all other methods have resulted in a large area of erroneous segmentation results, which compromises the quality of the generated segmentation map. The semantic segmentation results of images enhanced by our proposed method are closest to the ground truth, showing our performance significantly outperforming other state-of-the-art methods.

4.9. Advantages and limitations

4.9.1. Advantages

Compared with the exiting normalizing flow method, LLFlow [17] is the first to introduce normalizing flow into the LLIE tasks. However, it has two limitations: (1) LLFlow utilizes the super-resolution reconstruction module RRDB [56] as the feature extractor for low-light images. RRDB is not an optimal choice because it is based on CNN's residual-in-residual dense block, which only characterizes local features and fails to adequately consider the inherent characteristics of low-light images, resulting in insufficient ability to learn overall brightness and color, as well as redundant parameters (see Figs. 2, 14, 18, and Table 6). (2) The invertible network of LLFlow only adopts invertible network structure of SRFlow [11], which has limited capacity for image fitting. In contrast, our proposed UPT-Flow abandons RRDB and designs an unbalanced point-guide Transformer specifically for low-light images, and significantly improves the expression ability of

the flow network through the design of alternating cross-coupling and AATN in the invertible network of flow.

Compared to other methods, our approach does not rely on error loss or GAN loss. Methods based on these losses often result in blurry details, severe noise, and color distortion in the enhanced images (see Figs. 7, 9, and 11). UPT-Flow achieves high-quality image enhancement by directly calculating the probability density of the image. However, traditional normalizing flow-based methods (such as LLflow) struggle to accurately fit data distributions due to the limitations of their invertible computation. To address this, we designed the MSFormer and made improvements to the invertible network, which significantly enhance the fitting ability of the flow model to complex data like images, which is the key to improving the performance of our model.

4.9.2. Limitations

UPT-Flow mitigates the negative impact of distorted information on image enhancement by guiding the attention calculation of transformer through unbalanced point priors in low-light images. However, if the low-light image contains an excessive number of unbalanced points, these points will interfere with the attention calculation guided by the unbalanced points, leading to potential distortions in the enhanced images. As shown in Fig. 18, for images with large areas of unbalanced points, while our method exhibits significant advantages in detail processing, there is still some color distortion in the overall image (with a blue shift compared to the Ground Truth). Nonetheless, for such images with excessive distortion, most methods may struggle to yield satisfactory outcomes, which could be a focus for future research.

Moreover, according to Table 6, the inference time of UPT-Flow is longer compared to other methods, which may limit its applicability in real-time scenarios. Pruning and quantifying UPT-Flow could help its application in real-time scenarios.

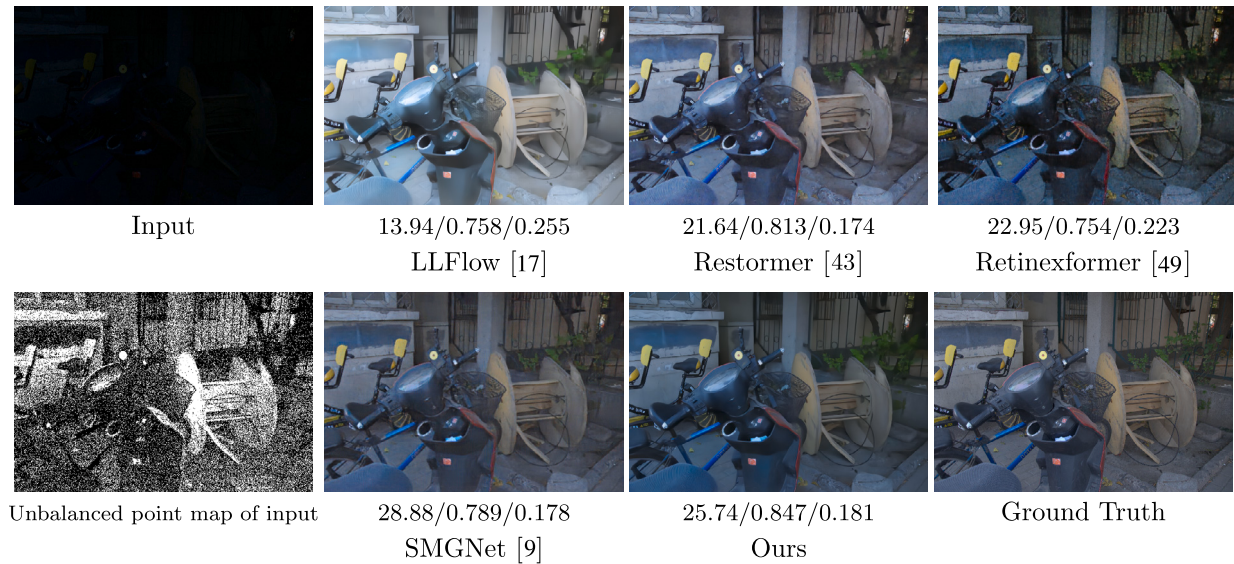


Fig. 18. Visual comparison of Limitations. The PSNR, SSIM, and LPIPS scores are listed below the enhanced image in order.

5. Conclusion

In this study, it is observed that low-light images often exhibit local information distortion due to unbalanced proportion RGB values. To address this, we propose an unbalanced point prior based on the unbalanced proportions of the RGB components in low-light images, which is employed to modify the attention computation, effectively reducing the impact of information distortion areas. Furthermore, we utilize global crossed-channel feature transformers to perform global information compensation in the window unbalanced points-guided transformers, thereby establishing a multi-scale Transformer with enriched global context and local fine-grained information for conditional normalizing flow. Lastly, we introduce cross-coupling condition affine layers based on channel and spatial attention within the invertible network of flow to enhance the expressive capability of a single flow step. Extensive experiments, including a large-scale user study, demonstrated the superior performance of our method on representative benchmarks.

Our model shows insufficient performance on images with a large amount of severe information distortion. In the future, we plan to explore other high-performance invertible components to enhance the nonlinear fitting capabilities of the flow network. Also, we plan to extend our method to other computer vision tasks such as image inpainting and image denoising.

CRedit authorship contribution statement

Lintao Xu: Writing – original draft, Visualization, Software, Methodology, Writing – review & editing. **Changhui Hu:** Supervision, Funding acquisition, Conceptualization. **Yin Hu:** Visualization, Validation. **Xiaoyuan Jing:** Validation, Data curation. **Ziyun Cai:** Validation, Formal analysis. **Xiaobo Lu:** Project administration, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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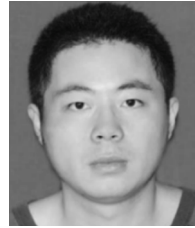
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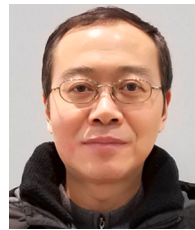
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